Below are **structured assignments** based on the **Education Domain tables** you previously created:

- Departments
- Students
- Teachers
- Courses
- Enrollments
- Exams
- Marks

Assignment Set – Views, Stored Procedures, and Functions (Education Domain)

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1. Views Assignments

Objective

Learn how to create **virtual tables to simplify complex queries**.

Assignment 1 – Student Department View

Create a view named:

vw_StudentDepartment

The view should show:

Column
StudentID
StudentName
DepartmentName
AdmissionDate

Tables used:

Students

Departments

Tasks

1. Create the view.
2. Retrieve all records from the view.
3. Filter students from **Computer Science department** using the view.
4. Drop the view.

Assignment 2 – Student Course Enrollment View

Create a view named:

vw_StudentCourses

The view should show:

Column
StudentName
CourseName
EnrollmentDate

Tables required:

Students

Enrollments

Courses

Tasks

Write queries using this view to:

1. Show courses taken by **StudentID = 5**
2. Count courses taken by each student
3. List students enrolled after **2024**

Assignment 3 – Exam Result View

Create a view:

vw_ExamResults

Columns:

StudentName

CourseName

ExamType

MarksObtained

Tables used:

Students

Marks

Exams

Courses

Tasks

1. Retrieve students scoring **more than 80**
2. Retrieve top scorers in each exam
3. Find students who **failed**

Assignment 4 – Aggregate View

Create a view:

vw_DepartmentStudentCount

Columns:

DepartmentName

TotalStudents

Tables used:

Students

Departments

Tasks

1. Retrieve departments with **more than 10 students**
2. Sort departments by highest student count.

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2. Stored Procedures Assignments

Objective

Learn how to encapsulate business logic inside **stored procedures**.

Assignment 1 – Insert Student Procedure

Create a stored procedure:

sp_InsertStudent

Parameters:

FirstName

LastName

Gender

DepartmentID

AdmissionDate

Tasks

1. Insert a new student.
2. Execute the procedure.
3. Verify inserted data.

Assignment 2 – Get Students By Department

Create procedure:

sp_GetStudentsByDepartment

Parameter:

@DepartmentID

Output:

StudentID

StudentName

AdmissionDate

Tasks

1. Execute procedure for **DepartmentID = 2**
2. Execute for **DepartmentID = 3**

Assignment 3 – Course Enrollment Procedure

Create procedure:

sp_EnrollStudent

Parameters:

@StudentID

@CourseID

The procedure should:

1. Insert data into **Enrollments table**
2. Automatically use **current date**

Assignment 4 – Student Marks Procedure

Create procedure:

sp_GetStudentMarks

Parameter:

@StudentID

Output should show:

StudentName

CourseName

ExamType

MarksObtained

Tables involved:

Students

Marks

Exams

Courses

Assignment 5 – Update Student Marks

Create procedure:

sp_UpdateMarks

Parameters:

@MarkID

@NewMarks

Tasks

1. Update student marks
2. Display updated result.

Assignment 6 – Delete Enrollment

Create procedure:

sp_DeleteEnrollment

Parameter:

@EnrollmentID

Tasks

1. Delete enrollment record.
2. Verify deletion.

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3. User Defined Functions Assignments

Objective

Learn how to create **scalar functions and table valued functions**.

Assignment 1 – Calculate Grade (Scalar Function)

Create function:

fn_GetGrade

Input:

MarksObtained

Output:

Grade

Logic:

Marks >= 90 → A

Marks >= 75 → B

Marks >= 60 → C

Else → Fail

Tasks

Use this function to display grades in exam results.

Assignment 2 – Student Age Function

Create function:

fn_GetStudentAge

Input:

DateOfBirth

Output:

Age

Tasks

Use function to display student age.

Assignment 3 – Total Marks Function

Create scalar function:

fn_GetTotalMarks

Input:

StudentID

Output:

TotalMarks

Tables used:

Marks

Assignment 4 – Student Courses Function (Table Valued)

Create function:

fn_GetStudentCourses

Input:

StudentID

Output:

CourseName

EnrollmentDate

Tables used:

Enrollments

Courses

Assignment 5 – Department Students Function (Table Valued)

Create function:

fn_GetDepartmentStudents

Input:

DepartmentID

Output:

StudentID

StudentName

AdmissionDate